Existence of Solutions Algebraic Equations from Existence of Solutions of Ordinary Differential Equations (常微分方程式の解の存在定理からの代数学の基本定理)

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Existence theorem of solutions of ordinary differential equations (Cauchy-Lipschitz theorem) follows without existence of algebraic eqations. We can proof Existence of solutions of algebraic equations from existence of solutions of ordinary differential equations. We consider complex coefficients. (常微分方程式の解の存在定理(コーシー-リプシッツの定理)の証明に代数学の基本定理は使わない. 代数学の基本定理を常微分方程式の解の存在定理から証明できる. 複素係数で考える.)

An initial value probrem $P(D)u(x)=0,\ x\in I, u^{(k)}(0)=u_{0k}\ (k=0,\dots,n-1)$ has an unique solution $u\in C^1(\bar{J})$ from corollary of Cauchy-Lipschitz theorem ([1]): if f is Lipschitz continuous on bounded and closed interval $I\ni 0$, then an initial value problem $u^{(n)}(x)=f(x,u,u',\dots,u^{(n-1)}),\ u^{(k)}(0)=u_{0k}\ (k=0,\dots,n-1)$ has an unique solution on bounded and closed interval $0\in J\subset I$.

Linear ordinary differential eqations with constant coefficients on bounded and closed interval I, an initial value problem $P(D)u(x)=0,\ x\in I, u^{(k)}(0)=u_{0k}\ (k=0,\dots,n-1)$ has an solution. To prove it, let
$0=P(x)=\sum_{i=0}^{n}a_i x^i\Rightarrow x^n=Q(x)=\sum_{i=0}^{n-1}-(a_i/a_n)x^i$.
Since $Q(x,x_1,\dots,x_{n-1})=\sum_{i=0}^{n-1}-(a_i/a_n)x_i^i\ (x_0=x)$ is Lipschitz continuous, so $u^{(n)}(x)=Q(D)u(x), u^{(k)}(0)=u_{0k}$ has a solution on the neighborhood $0\in J=\bar{J}\subset I$. To prove it, put $v=u^{(n-1)}, f(x,u,\dots,u^{(n-1)})=Q(D)u(x)$, use mathematical induction and Cauchy-Lipschitz theorem to $v'(x)=f(x,u,\dots,v), v(0)=u^{(n-1)}(0)$.

u has an zero-extension as a tempered distribution (not as a solution) $\bar{u} = u \, \mathrm{on} \, J, = 0 \, \mathrm{on} \, \mathbb{R} - J$. By Fourier transform as a tempered distribution, we obtain $P(i\xi)\mathcal{F}\bar{u}(\xi) = 0$ on J (There is a gap here). $P(i\xi)\mathcal{F}\bar{u}(\xi) = 0 \iff P(D)\bar{u}(x) = 0$ ([2]). $\exists \xi, \mathcal{F}\bar{u}(\xi) \neq 0$, so we have an existence of ξ such that $P(i\xi) = 0$.

References: [1]石原茂-矢野健太郎, 解析学概論, 裳華房, 2020; [2]垣田高雄-柴田良弘, ベクトル解析から流体へ, 日本評論社, 2024.